

SciOdyssey.

A research layer over your agent harness.

Our agent can carry a research task to completion.

PRESERVE THE RESEARCH WORLD.
RESET THE RESEARCHER.

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME
EVIDENCE

FRESH
JUDGMENT
A WIDER
SEARCH

Research
Runs
Longer.

01 / PROBLEM

Why current agents fall short.

Sustained autonomous exploration encounters three obstacles:

01 Closed-world workflow

Rigid predefined pipelines fail when encountering unexpected errors in open research.

02 Single trajectory inertia

Context accumulation anchors the agent to past reasoning, trapping search in local basins.

03 Context inflation cost

Continually rereading deep project context costs 8× more tokens than the actual code edits.

DESIGN RESPONSE ADAPTIVE TOOLS FRESH REASONING AUDITED EVIDENCE REUSABLE CONTEXT

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

01

Environment

Task contract · data
evaluator · baseline
budget · permissions

02

Research World

Verified ledger
Working priors
Retained State

03

Fresh Scientist

Hypotheses · code
experiments · reports
scientific judgment

04

META Review

Audit claims ↔ evidence
Scope conclusions
crystallize State

CONTEXT
RESET

fresh reasoning
no inertia



Selective persistence / Trajectory handoff

Audit → scope → preserve → fresh Scientist. Only verified findings enter durable memory.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · VERIFIED / EVIDENCE

Knowledge + ledger

Metrics, failure logs, and full report artifacts

B · CURATED / REVISABLE

Brief + working priors

Current understanding and hypotheses, open to revision

C · IMPLEMENTATION

Retained code + State

Reproducible pipeline version with clear provenance

CODING-AGENT HARNESS + RUNTIME Workspace isolation Runner + evaluator · Structured logging Artifact capture State operations

PARALLEL BREADTH Isolated branches → Post-hoc review → Explicit adoption into State

The next Scientist inherits experience, not prior reasoning momentum.

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000

Official FM baseline



0.6059363

SciOdyssey submission

+0.0043363 Primary improvement

GAUC 0.6728421
(+0.0054)

nDCG@5 0.5390304
(+0.0033)

46 interactions · 8-seed FM · CPU

02 SUSTAINED SEARCH

The search kept going.

0.6016310

C1: Valid baseline

0.6040901

C2: Expand features

0.6044289

C3: Reduce variance

0.6059363

C4: Retained recipe

4 autonomous cycles · 13 named experiments · 7 Full evaluations

Performance advanced systematically across distinct cycles rather than a single lucky edit.

03 AUTONOMOUS RECOVERY

Our agent is robust.

01 · PRESERVE

NumPy float32 broke
JSON serialization; logs
stayed intact.

02 · REPAIR

Scientist diagnosed
exception and patched
scalar serialization.

03 · RETAIN

Rerun succeeded cleanly;
META validated &
preserved State.

Autonomous traceback self-healing in Cycle 1; META audits proxy sampling fidelity and rejects spurious state promotions.

04 DIRECT BENCHMARK CONTROL

Beat Codex on this task.

0.6044533

Direct Codex Goal control

0.6059363

SciOdyssey retained

+0.0014830 Primary advantage

Direct single-agent baselines anchored in local tuning; SciOdyssey broke out via persistent world resets.

05 RESOURCE EFFICIENCY

Resource use is part of the evidence.

TOTAL PROJECT USAGE

~\$10

subscription-equivalent estimate

HARDWARE

1 × MacBook M2

All exploration & training on 1 laptop

SUBSCRIPTIONS

GPT Plus + Gemini Pro

7 days full autonomous research

48.240M

Total tokens (incl. cache reads)

4.020M

Non-cache input + output

0

GPU-hours (NumPy / CPU training)

All results verified on KuaiRand-Pure official validation split with complete reproducible artifacts.

04 / INSIGHTS

What the run taught us.

01 · DIVERSITY PRIOR

Different agents search differently.

Heterogeneous Scientists turn distinct search styles (local refinement vs. mechanism pivots) into a broader exploration prior.

02 · PARALLEL SYNTHESIS

Parallel search + review produces stronger candidates.

Independent Scientists explore in isolation; review selects winning branches for synthesis (observed +0.0040 gain).

03 · ENGINEERING PRACTICE

Branch isolation & unattended continuation.

Give each agent an isolated branch to prevent conflicts; use GitHub Actions to run, code, and commit overnight.

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Objectives · data inputs · constraints · evaluation protocol

PERSONAL.md

How to work

Preferences · research style · supervisory checkpoints

Define task → Single step → Autonomous run → Parallel explore → Inspect evidence

Autonomous by default.

Supervisable by design.



Coding agent harness

NATURAL LANGUAGE ENTRY



Use the research-agent skill.

Follow task.md and PERSONAL.md.

Let me review the results and evidence.